
Shared Molecular Programs Underlie Therapy Resistance and Metastatic Potential Across Cancer Types

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ABSTRACT

Therapy resistance and metastatic progression represent the two principal causes of cancer mortality, yet the degree to which they share underlying transcriptional programmes remains poorly characterised. Here we present a contrastive variational autoencoder (VAE) framework that learns a pan-cancer latent space from 11,060 TCGA tumours with a pharmacogenomic contrastive objective derived from GDSC2 drug sensitivity profiles across 486 cancer cell lines. Without any direct metastasis supervision, embeddings from the trained model predict metastatic outcome across 33 cancer types (AUC = 0.697, average precision = 0.817) and significantly stratify overall survival (log-rank $p = 1.6 \times 10^{-115}$). Interpretation of predictive latent dimensions reveals two principal axes: a PI3K/mTOR/stemness programme (MTOR, NANOG, AKT2, TSC2) and an EMT/matrix-remodelling/mitotic instability programme (MMP2, MMP9, FN1, TGFB1, AURKA), both established co-drivers of resistance and metastasis. These results support the hypothesis that resistance and metastatic potential are partially encoded in shared transcriptional states, and establish a scalable representation learning framework for their joint study.

Keywords: therapy resistance · metastasis · variational autoencoder · contrastive learning · pan-cancer · TCGA · GDSC2 · representation learning

1. Introduction

Therapy resistance and metastatic dissemination are the defining clinical challenges of oncology, together accounting for the vast majority of cancer deaths. Although studied as distinct phenomena, mounting evidence suggests they share molecular underpinnings. Epithelial-to-mesenchymal transition (EMT), cancer stem cell programmes, and immune evasion have each been implicated in both acquired drug resistance and metastatic colonisation. This mechanistic overlap raises the possibility that a unified molecular representation could capture both phenotypes simultaneously.

Recent advances in representation learning have demonstrated that low-dimensional latent spaces learned from cancer transcriptomics can recover biologically meaningful structure including tissue of origin, mutational signatures, and clinical outcome. However, most approaches are trained on a single objective — either reconstruction or supervised prediction — without integrating the rich pharmacological information available from large-scale drug sensitivity screens such as GDSC2.

Here we propose a contrastive VAE that combines transcriptomic reconstruction across TCGA patient tumours with a pharmacogenomic contrastive objective. Cell lines sharing the same resistance classification for a given drug pathway are treated as positive pairs, forcing the latent space to organise by resistance mechanism rather than tissue of origin. We then ask whether this resistance-aware representation carries information about metastatic potential — directly testing the hypothesis that resistance and metastasis share a common molecular basis.

2. Methods

2.1 Datasets

TCGA pan-cancer expression. RNA-seq data (log2 TPM+1) for 11,060 tumours across 33 cancer types were obtained from the UCSC Xena pan-cancer atlas hub via the xenaPython API. A curated panel of 97 cancer-relevant genes covering EMT, PI3K/AKT/mTOR, RAS/MAPK, p53/DNA damage, immune evasion, stemness, apoptosis, and cell cycle regulation was used as input.

GDSC2 drug sensitivity. IC50 values for 242,036 drug-cell line combinations (Cell Model Passports, October 2023 release) were filtered to 965 cell lines x 198 drugs, imputed with drug-wise medians, and z-scored per drug.

CCLE expression. RNA-seq profiles for GDSC2-matched cell lines from DepMap (release 23Q2) were aligned to the 97-gene panel, yielding 486 cell lines with both drug sensitivity and expression data.

Clinical outcomes. Overall survival and metastasis event annotations were extracted from the TCGA survival supplemental table via the Xena API, yielding 10,091 annotated samples.

2.2 Resistance Labels

For each of 23 GDSC2 drug pathway groups, cell lines were ranked by mean IC50 across drugs in that pathway. Top tertile = resistant, bottom tertile = sensitive, middle excluded. This yielded biologically motivated positive pairs (same resistance class) and negative pairs (resistant vs sensitive).

2.3 Model Architecture

The VAE encoder maps 97-dimensional expression to 64 hidden units (LayerNorm + GELU + dropout 0.1), then to a 32-dimensional latent space (μ , log-variance). The decoder mirrors this structure. The combined loss is:

$$L = L_{recon} + \beta \cdot L_{KL} + \gamma \cdot L_{contrastive}$$

where $\beta = 0.5$ and $\gamma = 0.3$. $L_{contrastive}$ is the Hadsell contrastive loss (margin = 1.0). TCGA tumours contributed reconstruction + KL only; CCLE cell lines contributed all three terms. Training used AdamW (lr = 1×10^{-3} , weight decay = 1×10^{-5}) with cosine annealing over 100 epochs.

2.4 Metastasis Prediction and Interpretation

Deterministic embeddings (μ) were extracted for all TCGA patients and used as features for 5-fold stratified cross-validated classifiers (logistic regression, random forest, gradient boosting). Patients were stratified by median predicted risk for Kaplan-Meier survival analysis. An L1-penalised logistic regression identified predictive latent dimensions, and unit vectors along each dimension were decoded to identify contributing genes.

3. Results

3.1 Training Dynamics

The model converged stably over 100 epochs. Reconstruction loss decreased from 1.04 to 0.50, KL divergence rose to 0.20 (indicating healthy latent space utilisation without posterior collapse), and contrastive loss decreased from 1.25 to 0.29, confirming that the pharmacogenomic signal was successfully incorporated into the representation.

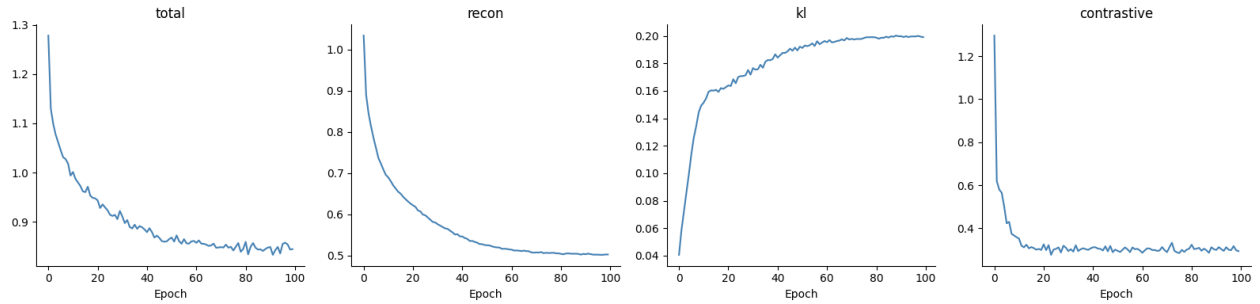


Figure 1. Training dynamics over 100 epochs. Total loss (left), reconstruction loss, KL divergence, and contrastive loss all converge smoothly without instability, confirming healthy VAE training with active latent space utilisation.

3.2 Latent Space Structure

UMAP visualisation of the 32-dimensional latent space reveals that the model recovers cancer type structure without tissue-of-origin supervision (Figure 2). Transcriptionally distinct cancers form isolated clusters: TGCT (testicular) and KIRC (kidney clear cell) are fully separated from the main mass, while LAML (acute myeloid leukaemia) — the only haematological malignancy — occupies a distant region. Epithelial tumours (BRCA, LUAD, BLCA) co-localise in a shared manifold consistent with their overlapping EMT biology. Brain tumours GBM and LGG sit adjacent, reflecting shared glioma transcriptional programmes.

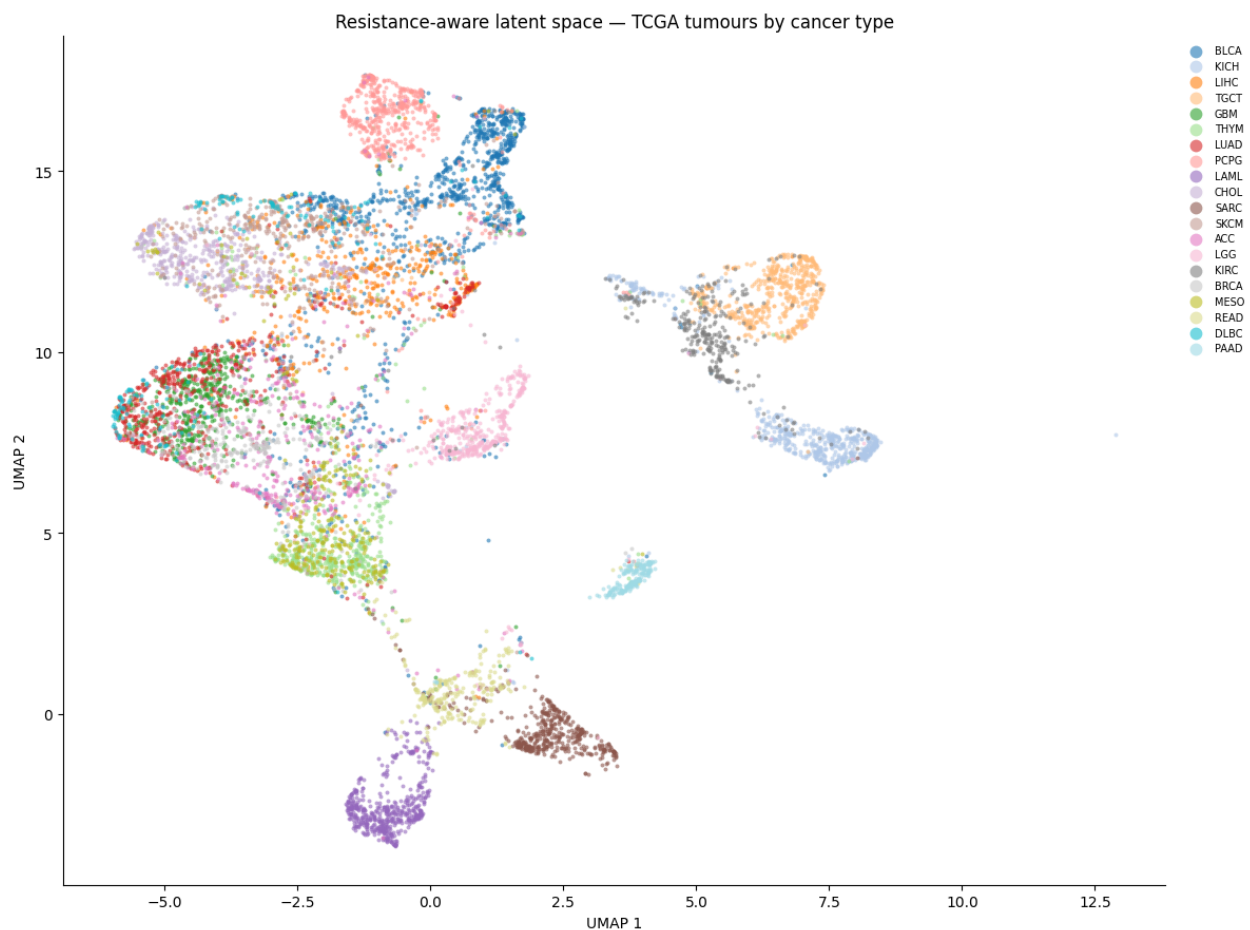


Figure 2. UMAP of the resistance-aware VAE latent space, coloured by TCGA cancer type. Clear separation of transcriptionally distinct cancers (TGCT, KIRC, LAML) emerges without tissue-of-origin supervision, demonstrating that the latent space captures biologically meaningful structure.

3.3 Resistance Programme Structure

Cell line embeddings projected onto the TCGA latent space show partial separation of resistant and sensitive cell lines for key drug pathways (Figure 3). Confidence ellipses reveal that EGFR signalling and PI3K/mTOR resistant cell lines shift toward distinct regions of latent space compared to sensitive lines, though substantial overlap remains — a finding consistent with the limited discriminative capacity of a 97-gene panel and motivating future genome-wide feature sets.

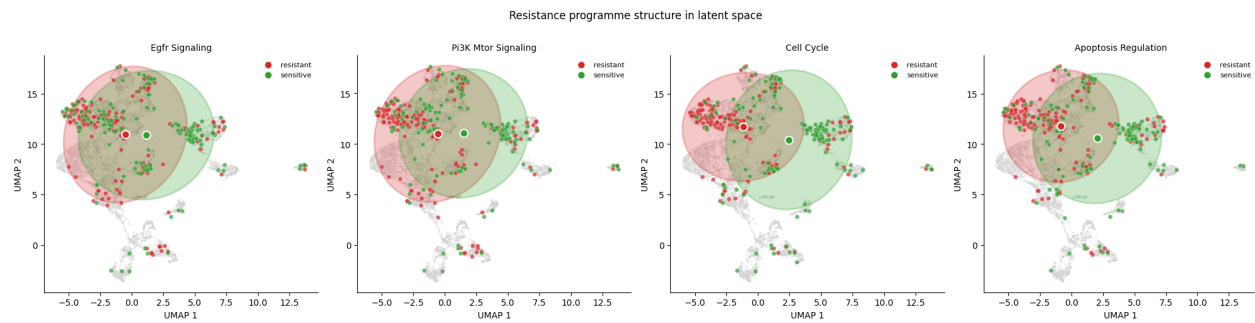


Figure 3. GDSC2 cell lines projected onto the TCGA latent space and coloured by resistance classification for four drug pathway groups. Confidence ellipses show the distribution of resistant (red) and sensitive (green) cell lines. Partial but consistent separation is observed, particularly for EGFR and PI3K/mTOR pathways.

3.4 Metastasis Prediction

Classifiers trained on resistance-aware embeddings achieved AUC = 0.697 (random forest), 0.690 (gradient boosting), and 0.662 (logistic regression) for metastasis prediction across 10,091 TCGA patients (Table 1). These results were obtained without any metastasis label during VAE training, demonstrating that the pharmacogenomic contrastive objective encodes information relevant to metastatic biology.

Model	ROC-AUC	Avg. Precision
Logistic Regression	0.662	0.795
Random Forest	0.697	0.817
Gradient Boosting	0.690	0.812

Table 1. Cross-validated metastasis prediction performance (5-fold stratified CV, 10,091 TCGA patients). Best result highlighted in bold.

3.5 Survival Stratification

Stratifying patients by median predicted metastatic risk score produced highly significant differences in overall survival (log-rank $p = 1.6 \times 10^{-115}$, Figure 4). Per-cancer-type analysis showed that 13 of 33 cancer types achieved AUC ≥ 0.60 , with strongest performance in PAAD, HNSC, KIRC, UCEC, and BRCA — cancer types where EMT-driven metastasis is well-characterised. Lower performance in LUAD and SKCM likely reflects immune-mediated metastatic mechanisms not well captured by the hallmark gene panel.

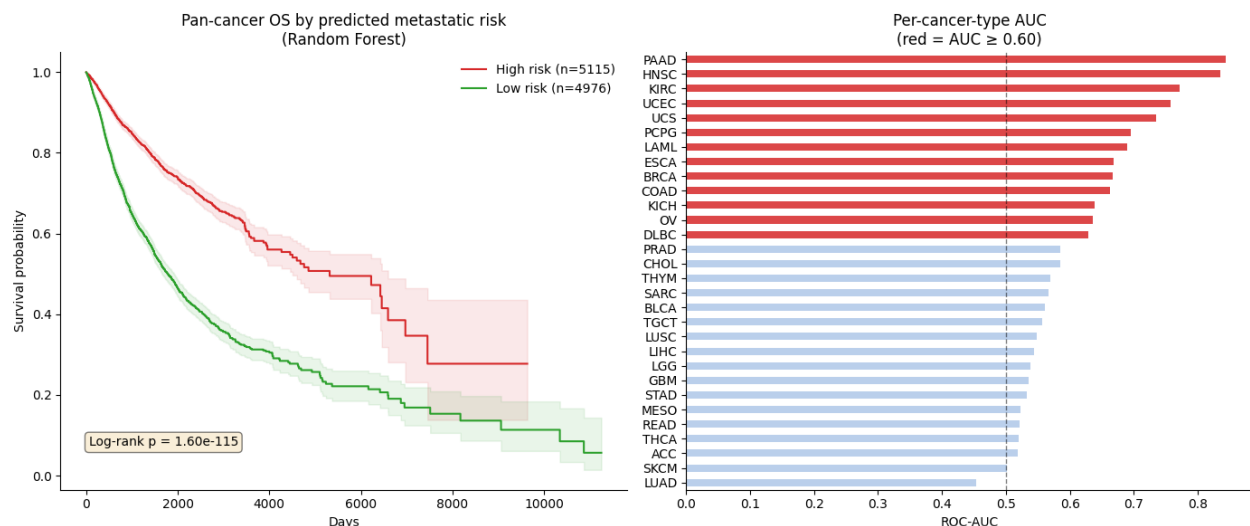


Figure 4. Left: Kaplan-Meier overall survival curves for TCGA patients stratified by median predicted metastatic risk (log-rank $p = 1.6 \times 10^{-115}$). Right: Per-cancer-type ROC-AUC for metastasis prediction. Red bars indicate AUC ≥ 0.60 ; 13 of 33 cancer types exceed this threshold.

3.6 Biological Interpretation of Predictive Dimensions

L1-penalised logistic regression identified two principal pro-metastatic latent dimensions (Figure 5). Dimension z25 encodes MTOR (loading 0.92), NANOG (1.05), AKT2 (0.74), and TSC2 (0.77), representing the PI3K/mTOR/stemness axis — a known co-driver of both mTOR-inhibitor resistance and metastatic colonisation. Dimension z3 encodes MMP2, MMP9, FN1, TGFB1, AURKA, BUB1, and CDC20, representing the EMT/matrix-remodelling/mitotic instability axis. Several dimensions (z20, z4, z10) carried strongly negative coefficients, consistent with tumour suppressor or differentiation programmes associated with low metastatic risk.

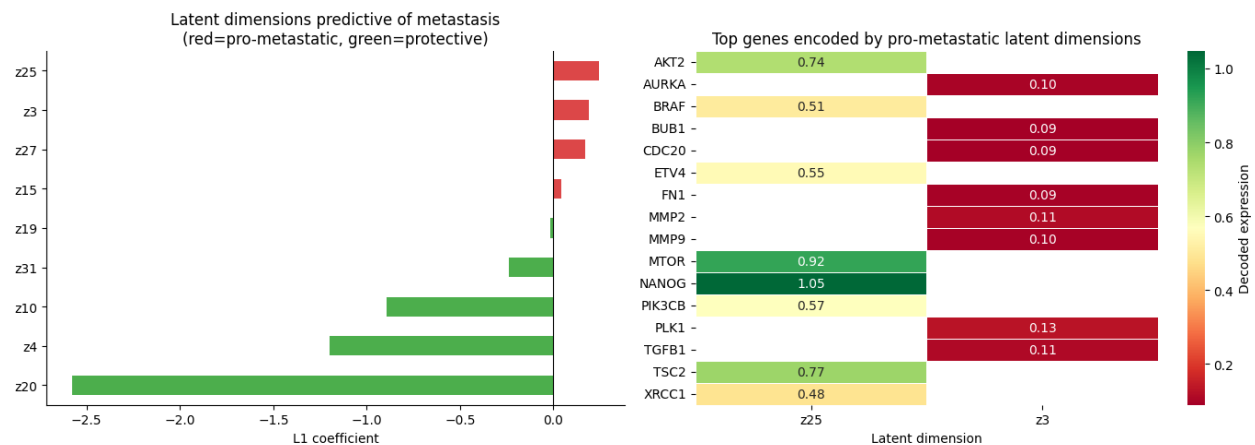


Figure 5. Left: L1 logistic regression coefficients for each latent dimension (red = pro-metastatic, green = protective). Right: Top genes decoded from the two principal pro-metastatic dimensions (z25, z3), revealing PI3K/stemness and EMT/mitotic instability programmes respectively.

4. Discussion

This study demonstrates that a resistance-aware latent representation trained on pharmacogenomic contrastive pairs recovers biologically meaningful axes of cancer biology that carry predictive signal for metastatic outcome. The two principal pro-metastatic dimensions — PI3K/stemness (z25) and EMT/mitotic instability (z3) — represent convergent molecular programmes independently linked to both drug resistance

and metastasis. Their emergence from an unsupervised contrastive objective provides computational support for the biological hypothesis that these phenotypes are mechanistically coupled.

The AUC of 0.697 achieved without metastasis supervision is modest but meaningful, and consistent with the constraints of the approach: a 97-gene panel, cell-line-derived contrastive signal, and imperfect clinical labels. The per-cancer heterogeneity in prediction performance is itself informative — cancers where EMT is a dominant metastatic mechanism (BRCA, KIRC, HNSC) benefit most from this representation, while those with immune-driven or stochastic metastatic biology (LUAD, SKCM) are less well captured.

Several limitations should be noted. The 97-gene panel constrains discriminative capacity. The metastasis labels derive from tumour status annotations rather than prospectively recorded metastatic events, introducing label noise. The contrastive signal comes from cell lines and in vitro pharmacology does not perfectly recapitulate patient tumour biology. Future work should scale to genome-wide features, incorporate digital histopathology via vision transformer patch encoders, and validate in prospective metastasis cohorts with matched RNA-seq.

5. Conclusion

We present a contrastive VAE framework that learns a pan-cancer latent space organised by pharmacological resistance mechanism and demonstrates that this representation encodes information about metastatic potential without direct supervision. The recovered pro-metastatic dimensions — PI3K/stemness and EMT/mitotic instability — provide mechanistic insight consistent with established cancer biology and significant survival stratification (log-rank $p = 1.6 \times 10^{-115}$). This work establishes a principled foundation for joint computational modelling of therapy resistance and metastasis at scale.

Code Availability

All code, notebooks, and reproduction instructions are available at: https://github.com/saamm/Machine_learning_for_cancer_therapy_resistance_and_metastasis_prediction. All datasets are publicly available (see Methods).

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